BOOST MWD

MUD PULSE MWD SYSTEM

Enedril

DRILLING

GENERAL DESCRIPTION

BOOST IS A POSITIVE MUD PULSE MWD SYSTEM THAT SENDS SURVEYS, DIRECTIONAL TELEMETRY, GAMMA RAY AND OTHER SENSOR DATA TO SURFACE IN REAL TIME, USING PRESSURE SIGNALS IN THE WELLBORE. THE CORE SYSTEM INCLUDES PULSERS, ELECTRONICS, SENSORS AND BATTERIES, WITH MANY CONFIGURABLE OPTIONS. ADJUSTABLE MODULE PLACEMENT ALLOWS ALTERNATIVE SENSOR POSITIONS, AND DOWN-LINKING ALLOWS ADJUSTABLE LOGGING SEQUENCES.

TECHNICAL CHARACTERISTICS

Outside diameter	47.625 mm (1.875")
Length, 1 battery	5.28 m (17.3 ft)
Length, 2 batteries	6.94 m (22.8 ft)
Pressure rating	137,895 kPa (20,000 psi)
Operating temperature	-20 to +185 °C (-4 to 347 °F)
High-temp. version	200 °C available
Rotary speed	60 rpm recommended
Wireline	Retrievable, re-insertable
Min. I.D. to retrieve	57 mm (2.25")
Bend radius	20° / 10 m
Collar sizes	89 - 203 mm (3.5" - 8")
Flow rate	0.28 - 4.45 m³/min (75 - 1,200 gpm)

Data transmission	Positive mud pulse
Power supply	Lithium 28 VDC / 3 A
Battery capacity	400+ h (2 battery modules)
Flow switch	Vibration, in pulser driver electronics
Shock limit	500 g, 0.5 ms, half sine, all axes
Vibration limit	5 - 30 Hz @ 1" d.a.; 30 - 500 Hz, 20 g
LCM tolerance	40 - 50 ppb, any size, premixed
Pulse width	0.250 - 2.00 s
Transmission time	10 - 28 s
Internal memory	24 MB
Programmable sequences	4
Top or bottom mounted	Both positive pulse versions

d.a.: double amplitude. Standard tools are rated to 185 °C; high-temperature options are available.

SENSOR SPECIFICATIONS

Parameter	Range	Accuracy	Spread
Inclination	0 - 180°	± 0.10°	0.20°
Azimuth	0 - 360°	± 0.50°	0.50°
Tool face	0 - 360°	± 0.20°	0.40°
Dip angle	0 - ± 90°	± 0.2°	0.40°
Magnetic field	0 - 100,000 nT	± 125 nT	250 nT
Temperature	-35 - 200 °C	± 0.5 °C	1.00

TELEMETRY PARAMETERS

Inclination, azimuth, magnetic, dip angle and gravity qualifiers, raw X, Y, Z survey, toolface, continuous inclination and azimuth, battery status, shock, vibration, RPM, stick-slip, temperature, generic variables, annular pressure, orbital gamma (360° available).

DESIGN FEATURES

- **Field-proven design:** rugged and dependable, for aggressive drilling conditions.
- **Anti-plugging pulser:** motor driven, performs well in LCM and high-solids mud.
- **Simple operation:** wide range of flow rates and rotary speeds.
- **Ratcheting military connector:** robust barrel-to-barrel inter-module link, easy to assemble and service.
- **Stable in the hole:** precision-trimmed centralizers; shock mitigation absorbs vibration, reduces wear and improves MTBF.
- **Custom configuration:** adjustable module placement, down-linking, configurable options.

HOW TO ORDER

Please specify: 1. collar size 2. top or bottom mounted version 3. 1 or 2 battery modules 4. temperature rating (185 °C standard, 200 °C option) 5. sensor options (e.g. orbital gamma) and logging sequences.

